Cell Seeding & Transfection

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SUSTech-BIO 2026 iGEM Team

Purpose

Seed adherent cells at a defined density and deliver plasmid DNA using Lipofectamine 3000 and P3000. Our 293T reference is 300,000 cells per 12-well plate well, usually reaching approximately 70% confluence after 18 h. The plate conversions and reagent recipes below provide starting conditions for systematic setup.

Materials & Equipment

Item	Supplier / identifier	Use / specification
HEK293T (293T) cells	Healthy adherent culture	Count viable cells after preparation using C01
Complete growth medium	Gibco DMEM + Cellmax FBS	Established 293T formulation in C01; use matching medium for other cell lines
Lipofectamine 3000 + P3000	Invitrogen; L3000001 / L3000008 / L3000015 [1]	Pack sizes: 0.1 / 0.75 / 1.5 mL lipid, respectively; supplied P3000; store at 4°C
Opti-MEM Reduced Serum Medium	Gibco	Use without added serum for complex preparation
Plasmid DNA	Sequence-verified preparation	Measured concentration; purified, low-endotoxin DNA suitable for transfection
Culture plates; sterile tubes and tips	Axygen tips; laboratory supplies	Flat-bottom, tissue-culture-treated plates; pipettes appropriate for calculated volumes
Cell counter; inverted microscope	Laboratory equipment	Viable-cell count, attachment and confluence checks
Cell culture cabinet; incubator	Laboratory equipment	Aseptic work; 37°C and 5% CO2 for routine 293T culture

Before You Start

Prepare a plate map with plasmid identities, total DNA per well, lipid dose and replicates. Include untransfected cells and an appropriate positive transfection control; include an empty-vector or other matched experimental control when the assay requires it.

Check healthy growth, morphology and attachment before plating. Record the plate manufacturer and growth area. The 18 h interval and 70% target describe our 293T workflow; other cell lines require separate optimization of seeding number, attachment time, confluence, DNA dose and lipid dose.

Prewarm the culture medium. Opti-MEM is used to prepare complexes, not to replace all of the growth medium. Keep medium composition consistent across groups. The product permits serum and antibiotics in the receiving culture medium; it does not require an automatic medium change after addition. [1]

Procedure

A. Calculate the seeding number

1. Use 300,000 viable 293T cells per well of a 12-well plate as the reference. For another format, calculate cells per well = 300,000 × (target well growth area / reference 12-well growth area).

The example below uses Corning reference areas: 3.8 cm^2 for a 12-well well. Replace both areas with the specifications of the plates actually used. These are area-scaled starting cell numbers, not measured outcomes for every format. [3]

Plate	Area / well (cm^2)	Factor vs. 12-well	293T cells to seed / well	Medium / well (mL)
6-well	9.5	2.50	750,000	2.0
12-well	3.8	1.00	300,000	1.0
24-well	1.9	0.50	150,000	0.50
48-well	0.95	0.25	75,000	0.25
96-well	0.32	0.084	Approximately 25,000	0.10

Medium volumes are practical starting volumes before adding complexes, taken from the transfection scaling guide. [2] Equal cell numbers per cm^2 do not guarantee equal confluence after 18 h; inspect each format before transfection.

2. Prepare a well-dispersed cell suspension using C01 and determine its viable-cell concentration C (cells/mL). Calculate suspension volume per well = required cells / C. Add medium to reach the selected plating volume; the cell suspension is included in this volume.

Example: at C = 1,000,000 viable cells/mL, a 12-well well needs 0.30 mL suspension plus 0.70 mL medium for 1.0 mL total. For multiple wells, prepare the required total plus a modest, explicitly calculated pipetting allowance.

B. Seed and check the culture

3. Remix the cell suspension gently before dispensing, as cells settle during plating. Seed the calculated number into each well, then distribute cells by gentle forward-backward and left-right movement. Avoid circular swirling that concentrates cells unevenly.

4. Incubate 293T cells at 37°C with 5% CO2 for approximately 18 h. Inspect several fields per well: cells should be attached, evenly distributed and approximately 70% confluent in our routine setup. Use actual cell condition to decide readiness; do not transfect solely because 18 h has elapsed.

For other cell lines, establish a small seeding-density pilot at the intended transfection time and select a density from observed attachment and confluence. The general manufacturer target is 70-90% at transfection. [1] Do not assume the 293T cell number or timing transfers directly.

C. Calculate DNA, Opti-MEM, P3000 and lipid

5. Set the total plasmid DNA mass D per well. For co-transfection, D is the sum of all plasmids. Calculate each plasmid volume as mass (µg) / stock concentration (µg/µL); if concentration is reported in ng/µL, divide it by 1,000 first. Keep total DNA comparable between experimental groups. [4]

Quantity	Calculation / selection
Each plasmid stock volume	V_i (µL) = m_i (µg) / c_i (µg/µL); total DNA volume = sum of V_i
P3000 volume	2 × D µL, where D is the numerical DNA mass in µg [1]
Lipofectamine 3000 volume	At the table DNA doses: 1.5 × D or 3 × D µL. Test the two doses in separate wells for a new setup. [1, 2]
Opti-MEM volume	Use the format-specific A and B volumes below. In this table, each equals 50 × D µL. This is a recipe relationship at the listed DNA doses, not a rule for unlimited DNA escalation. [2]

Use the following manufacturer starting recipes for one well. A is the DNA + P3000 solution; B is the lipid solution. Each row contains two alternative lipid doses, not two additions to the same well. [2]

Plate	Total DNA (µg)	Opti-MEM A + B (µL)	P3000 (µL)	Lipo 3000 low / high (µL)
6-well	2.5	125 + 125	5	3.75 / 7.5
12-well	1	50 + 50	2	1.5 / 3
24-well	0.5	25 + 25	1	0.75 / 1.5
48-well	0.25	12.5 + 12.5	0.5	0.375 / 0.75
96-well	0.1	5 + 5	0.2	0.15 / 0.3

The 48-well low dose is calculated as 0.375 µL (rounded to 0.37 µL in [2]). The 96-well recipe follows the published reagent table, while its seeding count follows exact area scaling; these two factors should not be conflated.

6. Calculate master mixes separately for each DNA composition and lipid dose. For n identical wells and an allowance f, multiply each component by n × (1 + f), then dispense one-well equivalents. A 10% allowance means f = 0.10; it increases preparation volume, not the dose per well. For submicrolitre components, use enough identical reactions to bring transfers within the pipette's reliable range.

D. Prepare A and B separately, then combine

7. Prepare solution A by adding the required plasmid stock(s) to the stated Opti-MEM volume, then adding P3000. Mix gently by pipetting. Prepare one DNA composition per tube or master mix. [1]

8. Prepare solution B in a separate tube: add the selected Lipofectamine 3000 dose to its stated Opti-MEM volume and mix. A brief 2-3 s vortex is specified for diluted lipid. [1] Do not add concentrated lipid directly to concentrated DNA.

9. Add solution A to solution B and gently pipette to mix without frothing. Combine the complete per-well quantities; incubate for 10-15 min at room temperature, then add to cells. [1] Keep this interval consistent across conditions.

Volume convention: the table lists Opti-MEM to ADD to each tube. DNA, P3000 and lipid contribute additional volume. Equal Opti-MEM portions therefore do not produce exactly equal final A/B volumes. Transfer the whole prepared per-well mixture, rather than removing a nominal 100 µL and leaving part of a 12-well dose behind.

Worked example: one 12-well well

Tube / component	Amount
Total DNA selected for this example	1 µg from a 0.5 µg/µL stock = 2 µL DNA
A: Opti-MEM + DNA + P3000	50 + 2 + 2 µL = 54 µL
B, low dose: Opti-MEM + lipid	50 + 1.5 µL = 51.5 µL
B, high dose: Opti-MEM + lipid	50 + 3 µL = 53 µL
Combined mixture for one well	105.5 µL (low) OR 107 µL (high)
Receiving growth medium	1.0 mL already in the well; final volume is approximately 1.106 or 1.107 mL

To compare both doses, use two separate wells, each receiving its own full 1 µg DNA dose. The example stock concentration is illustrative; recalculate DNA volume from the measured concentration. A two-plasmid mass ratio of 1:1 would allocate 0.5 µg to each, not 1 µg to each.

E. Add along the wall and mix the plate

10. Slowly add the prepared complexes dropwise along the inner wall of each well, allowing the liquid to enter the culture medium gently. Distribute additions along the accessible wall rather than delivering one forceful bolus. Keep the tip away from the cell layer; do not jet liquid onto the monolayer.

11. Immediately after addition, thoroughly but gently mix the plate in a cross pattern: move it forward and backward, then left and right, repeating until the liquid is evenly distributed. Keep the plate level enough to prevent spills. Do not substitute a single circular swirl, vigorous shaking or pipetting over the attached cells.

12. Return the plate to the incubator. Record addition time, actual confluence, DNA mass, stock volumes, reagent doses and any detachment. Inspect morphology the next day and use the time point defined by the downstream assay for the readout. Assess transfection together with cell health.

p.s.

The 300,000-cell reference means cells seeded, not an inferred cell count at 70% confluence. Record the observed confluence after 18 h; do not calculate a doubling time from this single reference.

Label tubes by contents: A = DNA + P3000; B = Lipofectamine 3000. The letters alone are not sufficient because other protocols may use the opposite convention.

Do not double the DNA dose when reading the manufacturer's two-condition optimization scheme. Each test well receives the same DNA mass and one selected lipid dose.

Keep the interval between mixing A/B and adding to cells consistent. Prepare manageable batches so later wells do not wait much longer than earlier wells.

Slow wall-directed addition protects the monolayer, but still requires immediate, thorough cross mixing to distribute the complexes throughout the medium.

Expected Outcome

In the routine 12-well 293T setup, expect approximately 70% confluence after 18 h, with attached cells distributed across the well. Area-scaled formats require visual confirmation. Successful delivery should produce the intended assay signal relative to controls while retaining acceptable cell morphology and survival; no fixed transfection-efficiency threshold is assumed.

Troubleshooting

Observed problem	Check first	Suggested action
Too sparse or crowded at 18 h	Viable-cell count, area conversion, attachment and growth state	Adjust the next seeding number from observed confluence. Let sparse, healthy cultures reach the intended state; reseed crowded cultures when needed. Optimize other cell lines separately.
Cells accumulate in patches	Clumps, settling during dispensing and plate movement	Remix the suspension between aliquots and use gentle cross movement after plating. Avoid circular swirling.
Cells detach during addition	Fast delivery, tip position and attachment before dosing	Slow the addition along the wall and keep the tip away from the monolayer. Use thorough but gentle cross mixing.
Low signal despite healthy cells	Positive control, DNA identity/concentration, A/B recipe and readout timing	Verify the delivered DNA dose and assay first. Compare the two lipid doses at fixed DNA mass; change one variable at a time. [4]
Rounding or death after transfection	Untreated controls, DNA quality and lipid/DNA dose	If controls remain healthy, evaluate a lower lipid dose and, separately, lower DNA input. Assess recovery after a gentle medium replacement when needed. [4]
Large well-to-well differences	Small-volume accuracy, mixing and complex incubation order	Prepare appropriate master mixes, retain one-well aliquot size and standardize addition timing and cross mixing.

Reference

[1] Thermo Fisher Scientific / Invitrogen. Lipofectamine 3000 Reagent Protocol. MAN0009872, Rev. C.0, 10 February 2016, pp. 1-2. Product handling and complex preparation. Accessed 2026-10-03. Open source

[2] Thermo Fisher Scientific / Invitrogen. Scaling Up or Down Lipofectamine 3000 Reagent Transfections. One-page scaling table; revision not stated. Accessed 2026-10-03. Open source

[3] Corning. Cell Culture Product Selection Guide, CLS-CC-010, pp. 11-12: growth areas of multiwell plates and flat-bottom 96-well microplates. Areas used as calculation examples, not confirmation of laboratory plate brand. Accessed 2026-10-03. Open source

[4] Thermo Fisher Scientific. Lipofectamine 3000 Transfection Reagent FAQs, product family L3000008: co-transfection, DNA quality, optimization and cytotoxicity. Accessed 2026-10-03. Open source